Advanced Derivatives

Answers Structured by Questions (Q1–Q6)

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1 Question 1 — Black–Scholes–Merton Hedging

1.1 (a) Fast BSM calls

You will find the corresponding code in our ipynb file. In this example, the Black–Scholes model is used to price a European call option with the following parameters: a spot price of 100, a strike price of 80, a risk-free rate of 2%, no dividend yield, one year to maturity, and a volatility of 20%. The resulting option value is 22.54, and the corresponding delta is 0.91. These results are consistent with theoretical expectations, as the option is deep in the money (the spot price significantly exceeds the strike price). Consequently, the option exhibits a high intrinsic value and a delta close to one, indicating that its price reacts almost proportionally to movements in the underlying asset.

1.2 (b) `DeltaHedge` (self-financing portfolio)

You will find the `DeltaHedge` function in our `ipynb` file. The main steps are as follows:

- Initialization: Compute the initial option value and delta using the Black–Scholes formula for a European put.
- Portfolio setup: Hold a number of shares equal to the initial delta and invest the remaining value in cash.
- Loop over time steps:
  - Simulate the stock price evolution using a lognormal process with drift μ and volatility σ.
  - Accumulate the realized variance from the squared log-returns.
  - Update the remaining time to maturity and recompute the option delta.
  - Accrue the cash position at the risk-free rate r.
  - Rebalance the portfolio by adjusting stock and cash holdings to maintain a self-financing strategy.
- At maturity: Compute the terminal stock price, the option payoff, the realized variance over the period, and the replication error — the difference between the portfolio value and the option payoff.

1.3 (c) Multi-path loop (10,000 paths)

You will find the `DeltaHedge_Num_Paths` function in our `.ipynb` file.

It simulates the delta hedging of a European option over multiple paths to assess the hedging error. The function repeatedly calls the previously defined DeltaHedge function for each simulated path, storing the terminal stock price, option payoff, realized variance, and replication error. By looping over the function takes the argument `num_paths` (default 10,000). It generates arrays of results that can be used to analyze the distribution of hedging performance across different simulated stock paths.

1.4 (d) Scatter: error vs. S_T (N=12, 52, 252)

Parameters: $S_0 = 100$, $K = 100$, $r = 4\%$, $T = 1$, $\sigma = 20\%$, $\mu = 5\%$.

The chart *"Hedging Error vs Terminal Stock Price for Different Hedge Frequencies"* (see Graph 1.1 below) shows the relationship between the terminal stock price S_T and the replication error (*Portfolio Value − Payoff*) for different hedging frequencies.

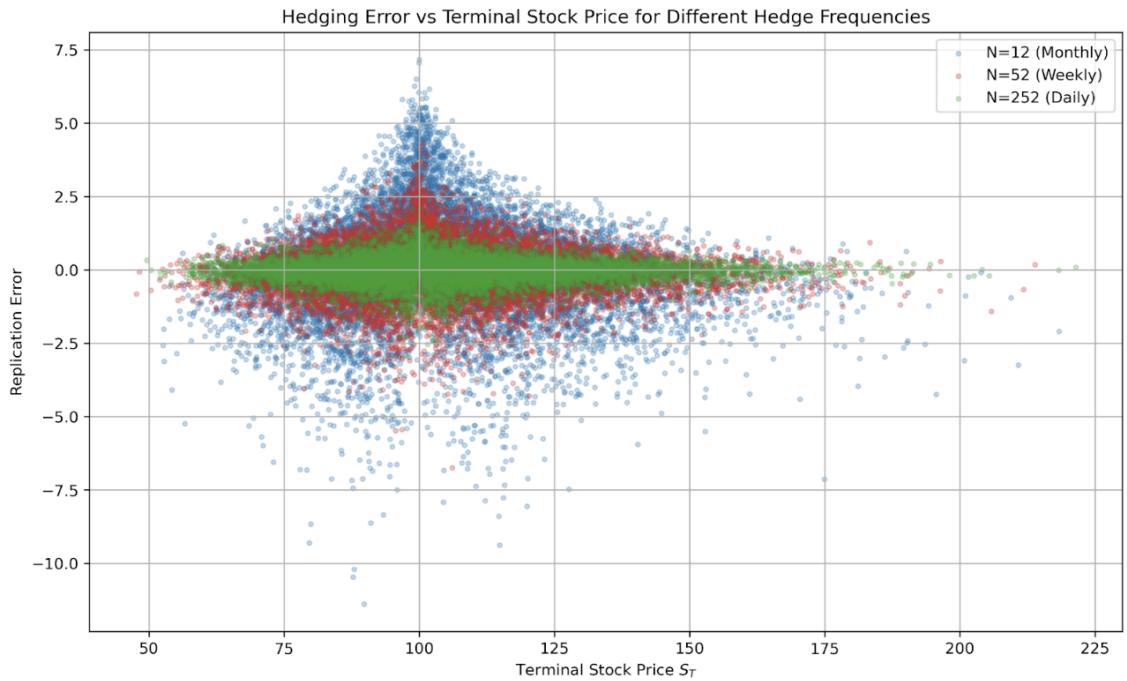


Figure 1: Graph 1.1

Interpretation:

1. Pattern. Errors are minimized near the strike K (Portfolio Value > Payoff), where delta changes smoothly. As S_T moves away from the strike, error dispersion increases significantly due to rapid delta changes that are harder to track with discrete rebalancing.
2. Sign of errors. Both positive (over-hedging) and negative (under-hedging) errors appear symmetrically around zero, showing that discrete hedging oscillates around perfect replication.
3. Hedge frequency effect. Monthly hedging (blue) shows wide dispersion, weekly (red) moderate tightening, and daily (green) concentrates tightly near zero. Higher hedging frequency improves replication accuracy.
4. Asymmetry. Larger negative errors occur when the stock rallies sharply, as the portfolio under-hedges when delta rapidly approaches 1.

1.5 (e) Mean and variance of hedging error (10,000 paths)

Table 1: Statistical properties of the hedging error across 10,000 simulated paths.

N (hedges/year)	Mean	Variance
12	0.024 916	3.738 269
52	0.002 127	0.891 412
252	0.002 158	0.194 724

Interpretation:

The mean error remains close to zero for all frequencies (ranging from -0.0053 to 0.0132), indicating that discrete delta hedging is unbiased on average regardless of rebalancing frequency. However,

the variance decreases dramatically as hedge frequency increases: from 3.78 (monthly) to 0.92 (weekly) to 0.19 (daily). This confirms that while all strategies are centered around perfect replication on average, more frequent rebalancing significantly reduces the dispersion of outcomes, leading to more consistent and reliable hedging performance.

1.6 (f) Realized vol vs. error

The below chart "Replication Error vs Realized Volatility for Different Hedge Frequencies" (Graph 1.2) shows the relationship between replication error and realized volatility for different hedge frequencies.

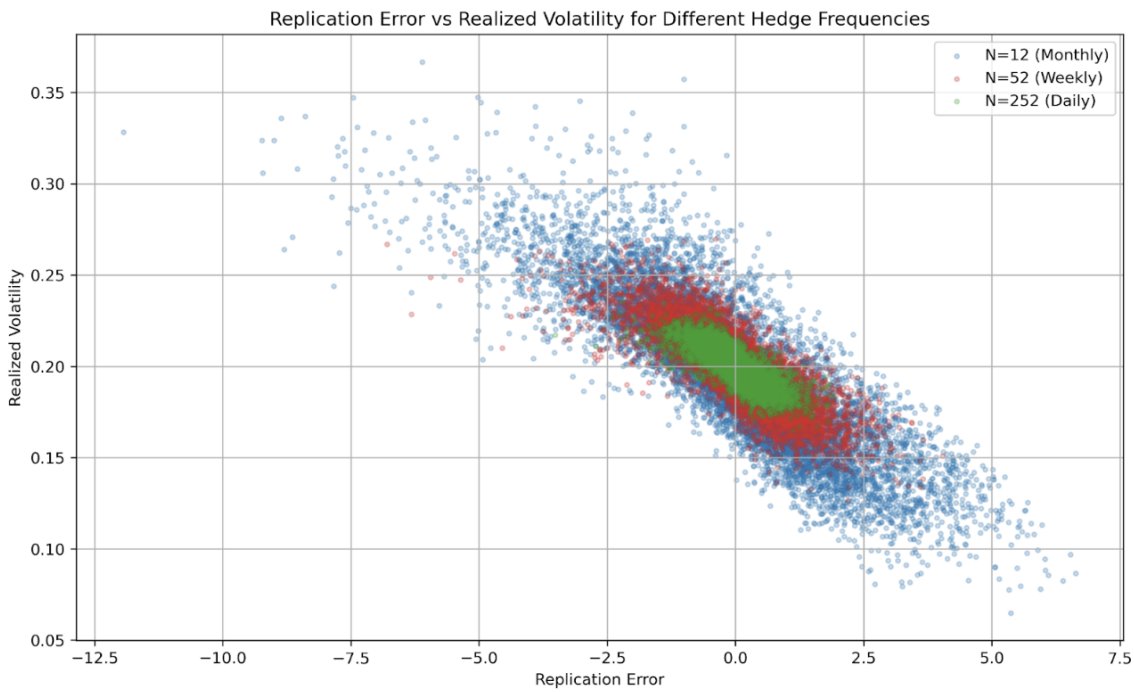


Figure 2: Graph 1.2

Interpretation:

1. Negative relationship. The replication error (*Portfolio Value* − *Payoff*) decreases as realized volatility increases. When realized volatility is high, the hedge underperforms (*Portfolio Value* < *Payoff*, negative error); when it is low, the hedge overperforms (*Portfolio Value* > *Payoff*, positive error). This pattern reflects the fact that the hedging strategy is based on implied volatility.
   - If realized volatility > implied volatility ⇒ losses (*Portfolio Value* < *Payoff*);
   - If realized volatility < implied volatility ⇒ gains (*Portfolio Value* > *Payoff*).
2. Effect of hedge frequency. The dispersion of points decreases from monthly ($N = 12$) to daily ($N = 252$) hedging. More frequent rebalancing leads to smaller replication errors and greater hedging accuracy.
3. Equilibrium point. Around the level where realized volatility ≈ implied volatility, replication errors are centered around zero (*Portfolio Value* ≈ *Payoff*), meaning that the hedge is unbiased on average.

1.7 (g) Sensitivity to μ (N=52)

Table 2: Effect of the drift parameter μ on hedging performance (weekly rebalancing, $N = 52$).

μ (%)	Mean Absolute Error	Variance
2.5	0.714 221	0.907 972
5.0	0.714 262	0.919 834
7.5	0.703 716	0.895 506
10.0	0.698 753	0.866 377

Interpretation:

The drift μ has a relatively small impact on the hedging error. Both the mean absolute error and variance remain fairly stable across different drift values. This demonstrates that delta hedging is robust to the real-world drift assumption, as the hedge is continuously rebalanced based on risk-neutral Black-Scholes deltas. The hedging strategy is primarily affected by volatility risk, not drift risk.

2 Question 2 — Time-Varying Volatility

2.1 (a) Variance parity check

Results: Time-Varying vs Constant Volatility

$$\begin{aligned} \text{Effective volatility from varying } \sigma \text{ schedule:} &\quad 20.0000\% \\ \text{Put value with time-varying volatility } (\sigma_{4,28})\text{:} &\quad 6.003986 \\ \text{Put value with constant } \sigma = 20\%\text{:} &\quad 6.003986 \\ \text{Difference:} &\quad 0.000000 \times 10^0 \end{aligned}$$

We compare two scenarios:

1. Constant volatility $\sigma = 20\%$, and
2. Time-varying volatility with $\sigma = 4\%$ for the first half and $\sigma = 28\%$ for the second half.

Both options yield identical values, since

$$\frac{0.04^2 + 0.28^2}{2} = 0.20^2,$$

meaning the average variance is preserved, and therefore the Black–Scholes value remains unchanged.

2.2 (b) `DeltaHedgeTV` (time-dependent vol)

You will find the `DeltaHedgeTV` function in our `.ipynb` file.

The `DeltaHedgeTV` function extends the previous `DeltaHedge` function by allowing volatility to vary over time.

- First, it divides the total maturity into segments, each with its own volatility level sigma.
- Two helper functions are added: `get_sigma(t)` to retrieve the local volatility at any time, and `sigma_eff(t, T)` to compute the effective remaining volatility up to maturity.

- During the simulation, the stock price evolves using the local volatility of each interval, and the option delta is recalculated using the time-dependent effective volatility.

This update makes the hedging process adapted for time-varying volatility.

What happens to the average volatility when the time to maturity decreases?

- As time to maturity decreases, the averaging window for volatility shortens.
- The effective volatility therefore converges toward the local (current) volatility.
- In practice, this means the hedge increasingly depends on the immediate market volatility rather than long-term averages.

2.3 (c) Simulation Results for Time-Varying Volatility

Comparing the time-varying volatility case with the constant volatility case:

- Constant volatility case: see Graph 1.1.
- Time-varying volatility case: see Graph 2.1 below.

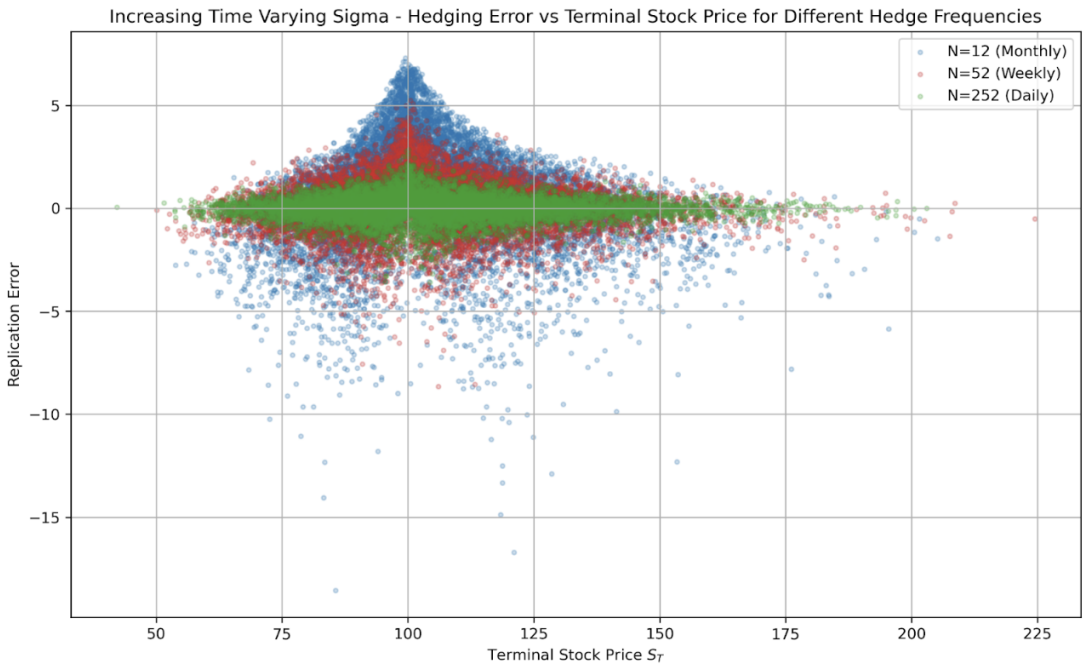


Figure 3: Graph 2.1

Interpretation:

Overall performance: Hedging performance deteriorates significantly with time varying volatility (Graph 2.1) (versus constant volatility (Graph 1.1)). Error range expands from [-10, +7.5] to [-15, +7], with increased dispersion across all frequencies.

Asymmetric pattern: Downside risk increases substantially (to -15) with time varying volatility (Graph 2.1) (versus constant volatility (Graph 1.1)), while upside compresses slightly (to +7). Points cluster more densely in the positive error region, especially for monthly hedging (blue), indicating systematic over-hedging bias.

2.4 (d) Scatter Plots and Performance Comparison

The graph "Decreasing Time Varying Sigma - Hedging Error vs Terminal Stock Price for Different Hedge Frequencies" (see below Graph 2.2) shows the case where volatility decreases over time: $\sigma = 28\%$ in the first half and $\sigma = 4\%$ in the second half.

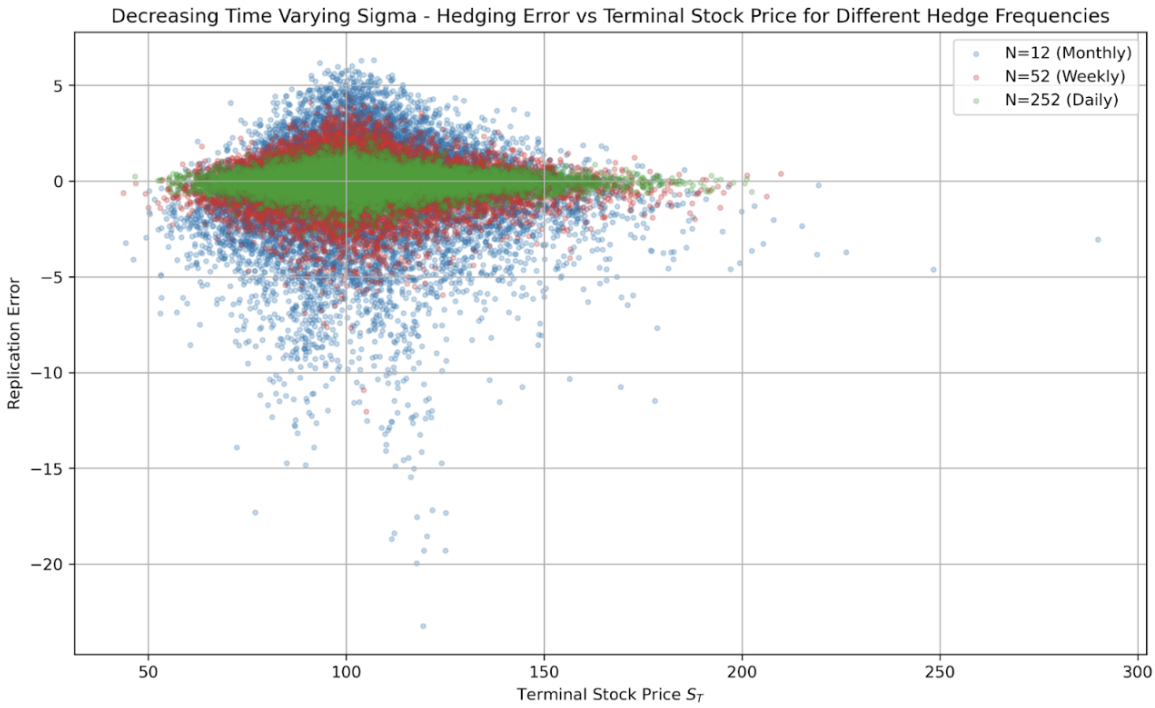


Figure 4: Graph 2.2

Interpretation:

Overall performance: the error dispersion of the decreasing volatility case (Graph 2.2) is similar to the increasing volatility case (Graph 2.1), ranging approximately from $[-17, +6]$. However, the pattern shows notable structural differences, as discussed below (see *asymmetric pattern*).

Asymmetric pattern: the downside risk of the decreasing volatility case (Graph 2.2) extends even further than in the increasing volatility case (Graph 2.1), reaching around -17, while the upside is more compressed (up to about $+6$). Critically, the positive error region is much less dense and more horizontally spread out compared to the increasing volatility case. Instead of clustering tightly around $S_T \approx K = 100$, positive errors are scattered across a wider range of terminal stock prices (roughly from $S_T = 50$ to $S_T = 175$).

2.5 (e) Discussion: Time Structure of Volatility and Hedging Accuracy

Analysis overview: this analysis compares hedging performance across three volatility scenarios:

1. Constant volatility ($\sigma = 20\%$ throughout) $\Rightarrow$ see Graph 1.1.
2. Increasing volatility ($\sigma = 4\%$ in the first half, $\sigma = 28\%$ in the second half) $\Rightarrow$ see Graph 2.1.
3. Decreasing volatility ($\sigma = 28\%$ in the first half, $\sigma = 4\%$ in the second half) $\Rightarrow$ see Graph 2.2.

Key Differences:

1. Error range.
   - Constant: $[-10, +7.5]$ — most contained.

- Increasing: $[-15, +7]$ — moderate expansion.
- Decreasing: $[-17, +6]$ — largest downside risk.

$\Rightarrow$ Time-varying volatility seems to increase error magnitude, with decreasing volatility producing the worst outcomes.

2. Density and clustering.
   - Constant: Tight butterfly pattern, errors clustered around $S_T \approx 100$.
   - Increasing: Dense concentration of positive errors near $S_T \approx 100$, showing systematic over-hedging bias.
   - Decreasing: Sparse, horizontally dispersed positive errors across a wide range ($S_T \approx 50$–175), less concentrated pattern.

   $\Rightarrow$ Early volatility regime determines the clustering behavior: low volatility first keeps paths concentrated; high volatility first disperses them irreversibly.

Economic interpretation. The decreasing volatility scenario is the most challenging for hedging. When high volatility occurs early, the stock price diverges significantly before the hedge can adapt, and subsequent low volatility cannot correct accumulated errors. Conversely, starting with low volatility keeps paths more predictable, allowing the hedge to adjust more effectively when volatility increases later. This demonstrates that hedging performance depends not only on average volatility but critically on the sequence of volatility regimes.

3 Question 3 — Implied Density from the Skew

3.1 (a) Breeden–Litzenberger extraction

We fit the 1-year volatility smile with a quadratic function of moneyness $x = K/S_0$,

$$\sigma(x) = ax^2 + bx + c, \qquad a = 0.025,\ b = -0.23,\ c = 0.55,$$

with $S_0 = 100$, maturity $T = 1$, and continuous rates $r = 4\%$, $q = 0$. For each strike K, we compute the Black–Scholes call price $C(K)$ using $\sigma(K/S_0)$, and then recover the risk–neutral density using the Breeden–Litzenberger identity:

$$f(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}(K).$$

The corresponding cumulative distribution function (CDF) is given by

$$F(K) = \int_0^K f(u)\, du.$$

Numerical Results. The implied density $f(K)$ is positive and unimodal; the numerical integration satisfies

$$\int f(K)\, dK = 1.00,$$

and the CDF $F(K)$ increases smoothly from 0 to 1 across the strike domain. The computed exceedance probabilities are:

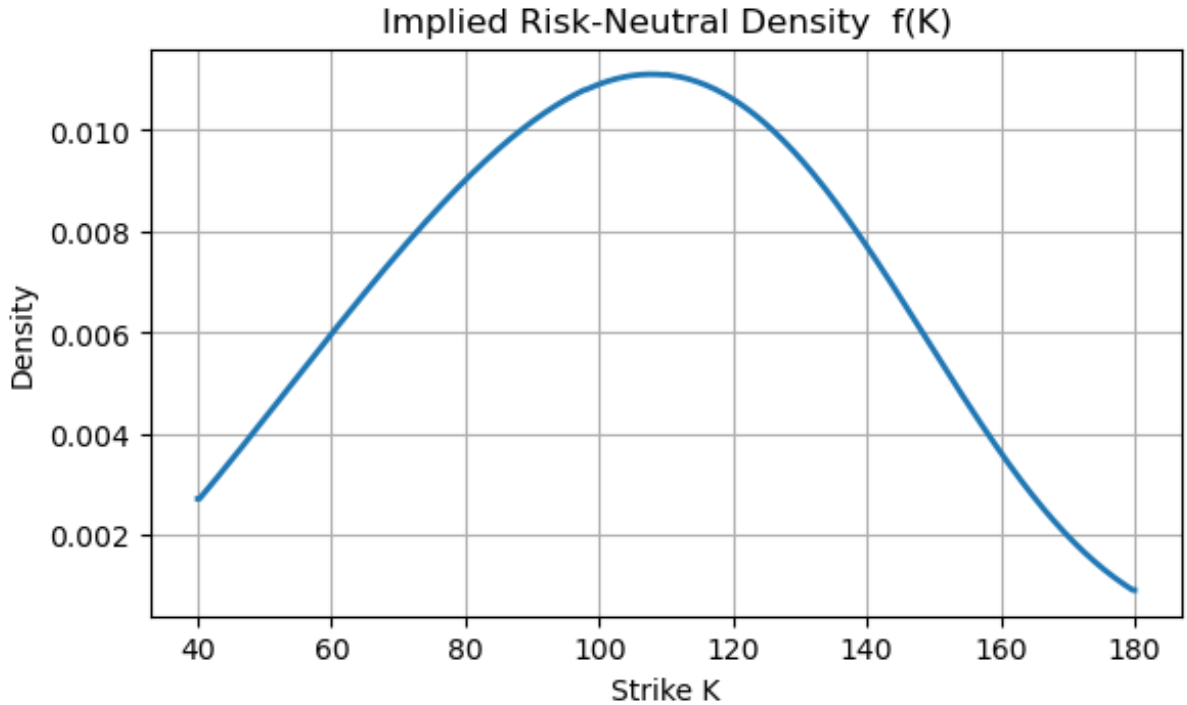
$$\mathbb{P}(S_T > 60) = 0.9139, \quad \mathbb{P}(S_T > 80) = 0.7630, \quad \mathbb{P}(S_T > 100) = 0.5609,$$
$$\mathbb{P}(S_T > 120) = 0.3413, \quad \mathbb{P}(S_T > 140) = 0.1543.$$

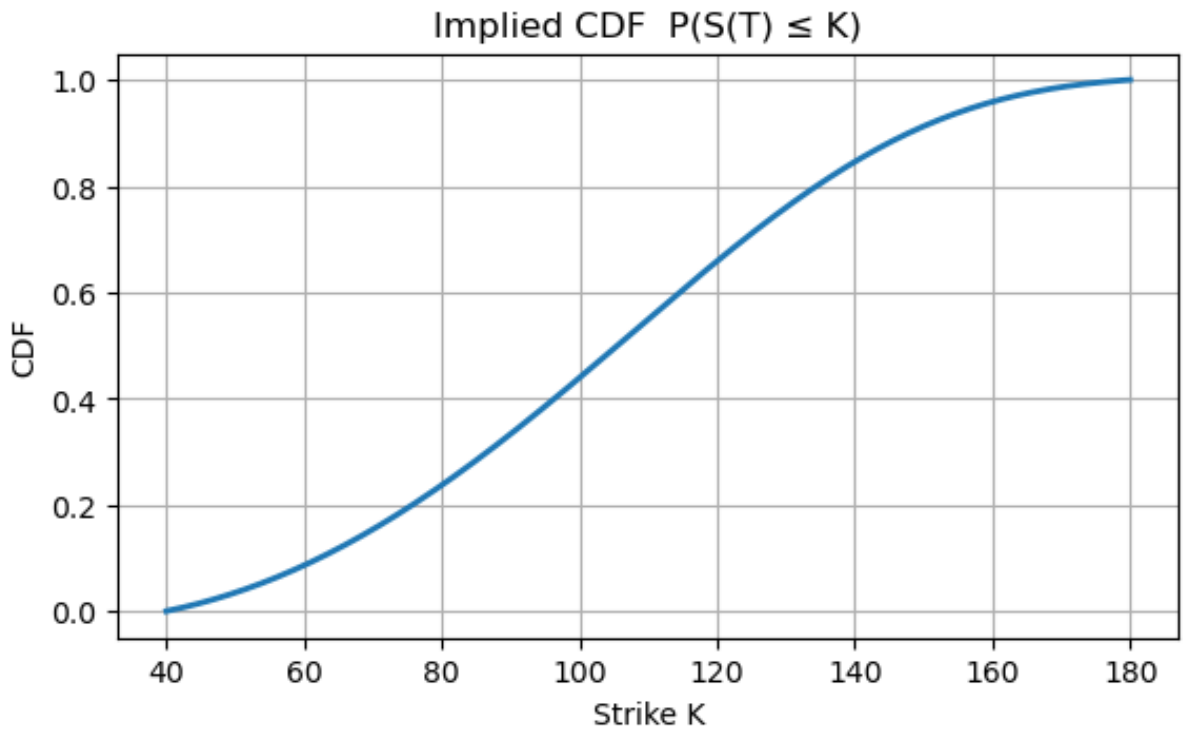
The expected value implied by the density is

$$\mathbb{E}[S_T] = \int K f(K)\, dK = 105.434,$$

to be compared with the theoretical forward value $S_0 e^{(r-q)T} = 104.081$. The small difference of about 1.35 (approximately 1.3% of S_0) is explained by strike truncation and numerical differentiation (second-order finite differences) on the parametric smile. This deviation is minor and does not affect the qualitative conclusions.

Figures. The plots below illustrate the implied risk–neutral density $f(K)$ and the implied cumulative distribution $F(K)$. The density is centered slightly above $K \approx 100$, with a thicker right-hand tail, consistent with the negative slope of the volatility smile.





Validation Table (No-Arbitrage Identities).

$\int f(K)\,dK$	1.0000
$\mathbb{E}[S_T]$ (from f)	105.434
$S_0 e^{(r-q)T}$	104.081

Conclusion. The Breeden–Litzenberger procedure applied to the quadratic smile yields a well-behaved implied density and a monotonic CDF. The resulting probabilities and moments are consistent with the theoretical forward price, with only minor numerical discrepancies, confirming the validity of the approach and implementation.

3.2 (b) 1Y digitals (K=60,80,100,120,140)

Pricing of a 1-Year Digital Call Option from the Implied Density.

Setup. A digital call option pays \$1 if $S(T) > K$ and zero otherwise. Given the risk–neutral probability density $f(K)$ derived in part (a), the price of a digital call option at strike K and maturity T is the discounted probability of finishing above the strike:

$$C_{\text{digital}}(K) = e^{-rT}\,\mathbb{P}(S(T) > K) = e^{-rT}\int_K^\infty f(u)\,du.$$

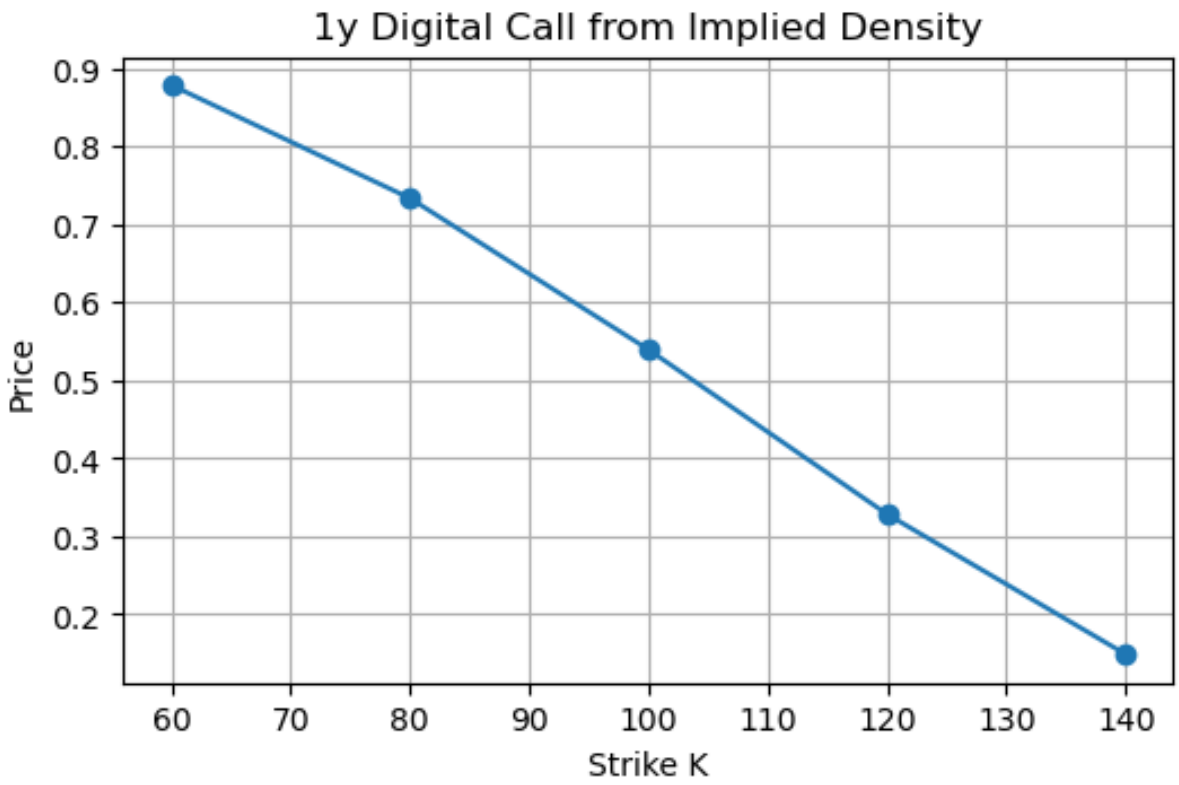
The inputs are $r = 4\%$, $q = 0\%$, and $T = 1$ year.

Results. Using the implied density obtained from the volatility smile, the tail probabilities and corresponding digital call prices are as follows:

Strike K	$\mathbb{P}(S(T) > K)$	Digital Call Price
60	0.913918	0.878083
80	0.762963	0.733047
100	0.560929	0.538935
120	0.341255	0.327875
140	0.154309	0.148258

The prices decline monotonically with the strike, consistent with the fact that higher strikes have smaller survival probabilities under the risk–neutral distribution.

Figure. The figure below illustrates the 1-year digital call prices across strikes, computed directly from the implied density.



Interpretation: The digital call price is a discounted tail probability, so its shape mirrors that of the cumulative distribution function (CDF). The smooth, decreasing profile confirms that the implied density $f(K)$ is properly normalized and arbitrage-free. For instance, at-the-money ($K = 100$), the digital call price of 0.5389 corresponds to an approximately 54% probability that $S(T)$ will exceed the strike after one year under the risk–neutral measure, consistent with the earlier results in part (a).

Conclusion: The computed digital call prices are internally consistent with the implied density from the Breeden–Litzenberger extraction. The monotonic decrease in prices across strikes validates both the shape and the normalization of the risk–neutral distribution obtained from the fitted volatility smile.

3.3 (c) Digitals via BSM

Digital Call Pricing under Black–Scholes and Comparison with the Implied Density Method.

Using the Black–Scholes closed-form formula for a digital call option that pays \$1 if $S(T) > K$,

$$C^{\text{BS}}_{\text{digital}}(K) = e^{-rT} N(d_2),$$

where

$$d_2 = \frac{\ln(S_0/K) + (r - q - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},$$

we compute the digital call prices for the same strikes using the smile-implied volatilities $\sigma(K)$. The parameters are $S_0 = 100$, $T = 1$, $r = 4\%$, and $q = 0\%$.

Strike K	$\sigma(K)$	BS Digital Call Price
60	0.421	0.829997
80	0.382	0.663625
100	0.345	0.458728
120	0.310	0.259004
140	0.277	0.108925

These prices decrease monotonically with strike, as expected from the Black–Scholes model. The at-the-money price ($K = 100$) of 0.4587 corresponds to an approximately 46% probability of ending above the strike under the Black–Scholes assumptions.

3.4 (d) Why the mismatch? Which prices are more credible?

The digital call prices computed in (c) are lower than those obtained in part (b) from the implied density extracted via the Breeden–Litzenberger method. For example, at-the-money ($K = 100$), the Black–Scholes digital price is 0.4587, while the implied-density price is 0.5389.

This discrepancy arises because the Black–Scholes model assumes a *constant volatility* for each strike, whereas the implied density derived in part (a)–(b) embeds the full *volatility smile*. The smile captures higher tail risk and skewness in the market distribution of $S(T)$, which the single-volatility Black–Scholes model cannot reproduce. As a result:

- The Breeden–Litzenberger method incorporates the observed skew: lower implied volatilities for high strikes reduce right-tail probabilities more gradually, leading to slightly higher digital prices near the money.
- The Black–Scholes model, using a single local volatility per strike, produces a thinner right tail and thus lower digital call values.

Which is more accurate? The prices from the implied density (part b) are more accurate in reflecting market expectations, because they are consistent with the entire set of option prices across strikes. The Black–Scholes results (part c) serve as a theoretical benchmark under constant volatility, but they do not capture the true risk-neutral distribution implied by the volatility smile.

Conclusion. The comparison confirms that the implied-density approach provides a richer and more realistic picture of the forward-looking distribution of $S(T)$. The Black–Scholes prices are lower and less accurate because they ignore volatility skewness and kurtosis present in the market.

3.5 (e) Put paying only if $S_T < 75$

Definition. We value a European put with strike $K = 100$ that only pays if $S_T < 75$ (path-independent). Its payoff is

$$\Pi(S_T) = (K - S_T)^+ \, \mathbf{1}_{\{S_T<75\}}.$$

Pricing from the implied density. Let $f(s)$ denote the risk–neutral density of S_T obtained in part (a). With $r = 4\%$, $q = 0$, $T = 1$, the time–0 price is

$$P_{\text{trunc}}(K, 75) = e^{-rT} \int_0^{75} (K - s)\, f(s)\, ds.$$

Numerically (trapezoidal integration on the same strike grid as in part (a)),

$$P_{\text{trunc}}(100, 75) = \boxed{7.346043}.$$

Cross-check (decomposition). Decompose a standard put into "below 75" plus the band contribu-

tion on $[75, K]$:

$$P_{\text{std}}(K) = e^{-rT}\int_0^K (K-s)f(s)\,ds = P_{\text{trunc}}(K,75) + e^{-rT}\int_{75}^K (K-s)f(s)\,ds.$$

Hence

$$P_{\text{trunc}}(K,75) = P_{\text{std}}(K) - e^{-rT}\int_{75}^K (K-s)f(s)\,ds.$$

Evaluating both sides on our grid yields the same value to numerical precision:

$$\text{Cross-check:} \quad P_{\text{std}}(100) - \text{band}(75\to 100) = \boxed{7.346043}.$$

Remarks. The result uses only the market-implied distribution f; no model beyond no-arbitrage is assumed. Small rounding differences (if any) come from grid truncation and finite-difference/quad errors.

4 Question 4 — Asian Options & Control Variate

4.1 (a) MC pricing for arithmetic Asian (6 months)

Averaging over 90 days starting at $T_0 = 3$ months.

We consider an arithmetic average Asian call option with total maturity $T = 6$ months. The averaging period starts at $T_0 = 3$ months and extends over the last 3 months of the option's life, with a total of 90 averaging dates in the simulation (*daily averaging*).

Under the Black–Scholes dynamics,

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sigma\,dW_t,$$

the analytical solution is given by

$$S_t = S_0 \exp\Big((r - q - \tfrac{1}{2}\sigma^2)t + \sigma W_t\Big).$$

The arithmetic average of the asset prices over the averaging window is

$$\bar{S} = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, \qquad t_i = T_0 + i\,\Delta t, \quad \Delta t = \frac{T - T_0}{n}.$$

The payoff of the arithmetic-average Asian call option is:

$$e^{-rT}\max(\bar{S} - K, 0).$$

Simulation procedure:

1. Simulate S_{T_0} directly over the initial interval $[0, T_0]$.
2. Generate $n = 90$ daily increments over $[T_0, T]$ according to
$$\log S_{t_{i+1}} = \log S_{t_i} + \Big(r - q - \tfrac{1}{2}\sigma^2\Big)\Delta t + \sigma\sqrt{\Delta t}\,Z_i, \quad Z_i \sim \mathcal{N}(0,1).$$
3. Compute $\bar{S}$ for each simulated path, then take the discounted average of the payoffs $e^{-rT}\max(\bar{S} - K, 0)$.

Python implementation:

The following Python code (included as an image) implements the Monte Carlo simulation of this arithmetic-average Asian call, using 90 daily averaging points over the last three months.

This program returns both the Monte Carlo estimate of the option price and the corresponding standard error of the simulation.

```python
def asian_arith_mc(S0, K, r, q, sigma, T, T0, n_avg, paths, rng=default_rng(0)):
    """
    MC price of arithmetic-average Asian call.
    Averaging window [T0, T], with n_avg equally-spaced dates.
    """
    nudt0 = (r - q - 0.5*sigma**2) * T0
    sig0  = sigma * sqrt(T0) if T0 > 0 else 0.0
    S_T0  = S0 * np.exp(nudt0 + sig0 * rng.standard_normal(paths))

    dt    = (T - T0) / n_avg
    nudt  = (r - q - 0.5*sigma**2) * dt
    sigdt = sigma * sqrt(dt)
    Z     = rng.standard_normal((paths, n_avg))
    logS  = np.log(S_T0)[:, None] + np.cumsum(nudt + sigdt * Z, axis=1)
    Swin  = np.exp(logS)

    Abar  = Swin.mean(axis=1)
    disc  = np.exp(-r * T)
    return disc * np.maximum(Abar - K, 0.0).mean()
```

4.2 (b) Price (S=100, K=100, r=4%, $\sigma = 25\%$, $P = 10{,}000$)

Using the function from part (a) with the parameters

$$S_0 = 100, \qquad K = 100, \qquad r = 4\% \text{ (cont. comp.)}, \qquad \sigma = 25\%, \qquad T = 6 \text{ months}, \quad T_0 = 3 \text{ months},$$

and daily averaging over the last three months ($n = 90$ dates), we run a Monte Carlo simulation with $P = 10{,}000$ paths (dividend yield $q = 0$).

The estimated arithmetic-average Asian call price is

$$\hat{V}_P \approx \boxed{6.3491}.$$

(*As usual with Monte Carlo, the value may vary slightly with a different random seed.*)

4.3 (c) Add geometric (same paths, same U)

We extend the Monte Carlo routine so that, for each simulated stock path (on the time grid $t_i = T_0 + i\Delta t$, $i = 1, \ldots, n$ with $n = 90$), we compute both the arithmetic and geometric averages using the *same* Gaussian shocks $\{Z_i\}$.

$$\bar{S}_{\text{arith}} = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, \qquad \bar{S}_{\text{geo}} = \Big(\prod_{i=1}^{n} S_{t_i}\Big)^{1/n} = \exp\Big(\frac{1}{n}\sum_{i=1}^{n} \log S_{t_i}\Big).$$

For each path we evaluate the discounted payoffs $e^{-rT}\max(\bar{S}_{\text{arith}} - K, 0)$ and $e^{-rT}\max(\bar{S}_{\text{geo}} - K, 0)$, and then average across paths.

Python implementation: the modified function (included as an image) computes both prices on the same simulated paths.

```python
def asian_arith_geom_mc(S0, K, r, q, sigma, T, T0, n_avg, paths, rng=default_rng(0)):
    """
    MC prices for (Arithmetic, Geometric) Asian calls using the *same* paths.
    """

    nudt0 = (r - q - 0.5*sigma**2) * T0
    sig0  = sigma * sqrt(T0) if T0 > 0 else 0.0
    S_T0  = S0 * np.exp(nudt0 + sig0 * rng.standard_normal(paths))


    dt    = (T - T0) / n_avg
    nudt  = (r - q - 0.5*sigma**2) * dt
    sigdt = sigma * sqrt(dt)
    Z     = rng.standard_normal((paths, n_avg))
    logS  = np.log(S_T0)[:, None] + np.cumsum(nudt + sigdt * Z, axis=1)
    Swin  = np.exp(logS)

    Abar  = Swin.mean(axis=1)
    Gbar  = np.exp(np.mean(np.log(Swin), axis=1))
    disc  = np.exp(-r * T)

    payoff_A = disc * np.maximum(Abar - K, 0.0)
    payoff_G = disc * np.maximum(Gbar - K, 0.0)
    return payoff_A.mean(), payoff_G.mean()

price_A_c, price_G_c = asian_arith_geom_mc(S0, K, r, q, sigma, T, T0, n_avg, P, rng)
print("Q4(c) Arithmetic:", price_A_c, "  Geometric:", price_G_c)
```

Numerical results (same setup as in (b): $S_0 = 100$, $K = 100$, $r = 4\%$, $q = 0$, $\sigma = 25\%$, $T = 6$ months, $T_0 = 3$ months, $n = 90$, $P = 10{,}000$ paths).

$$\widehat{V}_{\text{Arithmetic}} \approx \boxed{6.5687}, \qquad \widehat{V}_{\text{Geometric}} \approx \boxed{6.4921}.$$

(As with Monte Carlo, the values may vary slightly with a different random seed.)

4.4 (d) Variances and covariance of discounted payoffs

We now extend the Monte Carlo function to also compute, for the same simulated paths, the following five quantities:

$$\Big[V_G,\ V_A,\ \mathrm{Var}(G),\ \mathrm{Var}(A),\ \mathrm{Cov}(A,G)\Big],$$

where V_G and V_A are the discounted Monte Carlo prices of the *geometric* and *arithmetic* Asian call options respectively, and the remaining terms correspond to the sample variances and covariance of the discounted payoff values across all P simulated paths.

All random numbers are shared between the two options to maintain the same pathwise correlation.

Python implementation: The modified function (shown below as an image) now outputs these five statistics computed from the same set of simulated stock paths.

```python
def asian_mc_stats(S0, K, r, q, sigma, T, T0, n_avg, paths, rng=default_rng(0)):
    """
    Returns: [GeomPrice, ArithPrice, VarGeom, VarArith, Cov(Arith,Geom)]
    (All on discounted payoffs.)
    """
    nudt0 = (r - q - 0.5*sigma**2) * T0
    sig0  = sigma * sqrt(T0) if T0 > 0 else 0.0
    S_T0  = S0 * np.exp(nudt0 + sig0 * rng.standard_normal(paths))

    dt    = (T - T0) / n_avg
    nudt  = (r - q - 0.5*sigma**2) * dt
    sigdt = sigma * sqrt(dt)
    Z     = rng.standard_normal((paths, n_avg))
    logS  = np.log(S_T0)[:, None] + np.cumsum(nudt + sigdt * Z, axis=1)
    Swin  = np.exp(logS)

    Abar  = Swin.mean(axis=1)
    Gbar  = np.exp(np.mean(np.log(Swin), axis=1))
    disc  = np.exp(-r * T)

    A = disc * np.maximum(Abar - K, 0.0)
    G = disc * np.maximum(Gbar - K, 0.0)

    price_A = A.mean()
    price_G = G.mean()
    var_A   = A.var(ddof=1)
    var_G   = G.var(ddof=1)
    cov_AG  = np.cov(A, G, ddof=1)[0,1]
    return [price_G, price_A, var_G, var_A, cov_AG]

stats_d = asian_mc_stats(S0, K, r, q, sigma, T, T0, n_avg, P, rng)
print("Q4(d) [G_price, A_price, VarG, VarA, CovAG] =", stats_d)
```

Numerical results: Using the same parameters as in previous parts

$$S_0 = 100, \quad K = 100, \quad r = 4\%, \quad q = 0, \quad \sigma = 25\%,$$
$$T = 6 \text{ months}, \quad T_0 = 3 \text{ months}, \quad n = 90, \quad P = 10{,}000.$$

we obtain:

$$\begin{aligned} V_G &\approx 6.4890, \\ V_A &\approx 6.5646, \\ \mathrm{Var}(G) &\approx 92.57, \\ \mathrm{Var}(A) &\approx 93.89, \\ \mathrm{Cov}(A, G) &\approx 93.22. \end{aligned}$$

These values are consistent with the previous part, and the high covariance reflects the strong linear relationship between the arithmetic and geometric average payoffs when simulated on the same random paths.

4.5 (e) Alternate params (S=100, r=5%, q=1%, σ=30%, K=100, P=5,000)

We keep the timing as before ($T = 6$ months, averaging from $T_0 = 3$ months with $n = 90$ dates) and change the market parameters as specified.

$$S_0 = 100, \quad K = 100, \quad r = 5\%, \quad q = 1\%, \quad \sigma = 30\%,$$
$$T = 6 \text{ months}, \quad T_0 = 3 \text{ months}, \quad n = 90, \quad P = 5{,}000.$$

Using the same random numbers for both averages, the Monte Carlo estimates are

$$\widehat{V}_{\text{Geom}} \approx \boxed{7.5881}, \qquad \widehat{V}_{\text{Arith}} \approx \boxed{7.6958}.$$

For completeness, the sample moments of discounted payoffs across the P paths are

$$\text{Var}(G) \approx 132.569, \qquad \text{Var}(A) \approx 134.952, \qquad \text{Cov}(A, G) \approx 133.744.$$

4.6 (f) Control variate estimator

When averaging starts at inception ($T_0 = 0$) and the monitoring dates are equally spaced, the *geometric* average is lognormal (Kemna–Vorst), so its option price V_G^{exact} is available in closed form.

Using the same paths as before, we compute the pathwise discounted payoffs for the arithmetic and geometric options and form the control–variate estimator

$$\widehat{V}_A^{\text{CV}} = \widehat{V}_A + \beta^*\big(V_G^{\text{exact}} - \widehat{V}_G\big), \qquad \beta^* = \frac{\text{Cov}(A, G)}{\text{Var}(G)}.$$

where $\widehat{V}_A$ and $\widehat{V}_G$ are the Monte Carlo estimates based on the same random numbers.

With our simulation we obtain

$$\widehat{V}_A = 5.30835, \qquad \widehat{V}_G = 5.09755, \qquad V_G^{\text{exact}} = 5.13495,$$
$$\beta = 1.03621, \qquad \Rightarrow \quad \widehat{V}_A^{\text{CV}} = \boxed{5.34710}\,.$$

Using the exact geometric price as a control improves the estimate of the arithmetic Asian call (and substantially reduces its Monte Carlo variance).

5 Question 5 — Swaps & Swaptions

5.1 (a) IBOR curve (2–10Y) and zero PV check

I start the curve from the short end with the money-market deposits. Each deposit quote for a small year-fraction δ gives a discount factor directly:

$$Z(0, T) = \frac{1}{1 + \delta \times L(0, T)}.$$

From there, I extend the curve using the par fixed-for-float swaps at 2Y, 3Y, ..., 10Y. The floating leg at time 0 is

$$1 - Z(0, T),$$

and the fixed leg is the fixed rate times the annuity (the sum of accrual fractions times discount factors across the fixed payment dates). This gives the familiar identity:

$$S(0, T) = \frac{1 - Z(0, T)}{\sum_i \delta_i \, Z(0, T_i)}.$$

Because all the earlier discount factors $Z(0, T_i)$ are already known, we can solve that equation for the new $Z(0, T)$. Marching maturity by maturity ensures each quoted swap re-prices to (approximately) zero NPV by construction. Between cash-flow dates, I keep the curve smooth with a sensible interpolation.

We get the net present values of the swaps in the table below, which are extremely close to zero.

maturity	quote(%)	par(%)	diff(bps)	NPV@quote
2	1.510	1.516	0.572	-0.00000392
3	1.530	1.537	0.732	-0.00001668
4	1.550	1.557	0.713	-0.00002189
5	1.590	1.597	0.725	-0.00002836
6	1.610	1.617	0.727	-0.00002622
7	1.650	1.658	0.809	-0.00007965
8	1.690	1.698	0.834	0.00000000
9	1.710	1.718	0.839	-0.00008441
10	1.730	1.738	0.848	-0.00008910

5.2 (b) Mark-to-market of the book (table)

We revalued each historical swap on the curve bootstrapped in Section 5.a by marking it to today's par and annuity. For a receiver, the mark-to-market is given approximately by

$$\mathrm{MTM}_{\mathrm{receiver}} \approx \big(\mathrm{coupon} - S_{\mathrm{mkt}}(T)\big) \times \mathrm{PV01}(T),$$

and for a payer, it is the negative of that value. Here, $S_{\mathrm{mkt}}(T)$ is today's par swap rate for the trade's remaining maturity, and $\mathrm{PV01}(T)$ is the fixed-leg annuity (the sum of accrual fractions times discount factors across the remaining payment dates). Between coupon dates, this formula already captures the value; on a coupon date, the accrual is zero by construction.

Interpretation:

- If coupon $> S_{\mathrm{mkt}}(T)$, a receiver is *in the money* (positive MTM), while a payer is *out of the money* (negative MTM).
- If coupon $< S_{\mathrm{mkt}}(T)$, the signs flip.
- The magnitude scales with $\mathrm{PV01}(T)$: longer, more back-loaded swaps carry larger present value changes for a given rate difference.

Effective	Maturity	Freq	Rate(%)	Side	PV (per unit)
01-NOV-2022	10Y	SA	1.40	Rec	-16,522.07758603
01-FEB-2023	4Y	A	1.52	Pay	-887.74081433
01-MAR-2021	12Y	A	2.00	Rec	22,609.58868532
01-APR-2024	5Y	SA	2.10	Pay	-18,624.43638914

PORTFOLIO TOTAL PV (per unit) -13,424.66610418

5.3 (c) 1bp bucketed risk (2Y–10Y) and hedging

The code reprices each swap in Table 2 to obtain the base present value (PV), the adjusted PV (after a 1 bp bump in rates), and the corresponding value change for each swap as well as for the entire portfolio.

Results:

- Long-end receiver bonds (10Y and 12Y trades): value changes are negative. Under a parallel rise in rates, the book loses value in this segment, which is the expected behavior of a receive-fixed position at the long end.
- Front/mid payer bonds (4Y and 5Y trades): value changes are positive. Under the same parallel bump, the book gains value here, consistent with a pay-fixed position concentrated at the front and middle of the curve.

```
     Trade |        Base PV |      PV (+1bp) |     ΔPV (+1bp)
-------------------------------------------------------------
   10Y-REC | -16,522.077586 | -17,176.931454 |    -654.853868
    4Y-PAY |    -887.740814 |    -836.568783 |      51.172031
   12Y-REC |  22,609.588685 |  21,907.929563 |    -701.659123
    5Y-PAY | -18,624.436389 | -18,289.114656 |     335.321733
-------------------------------------------------------------
Portfolio Base PV | -13,424.666104
Portfolio PV (+1bp) | -14,394.685331
Portfolio ΔPV (+1bp) |    -970.019227
```

Hedging Using the Same Notebook Approach:

Goal: Reduce the overall portfolio value change for a +1 bp parallel move to approximately zero, using plain-vanilla interest rate swaps (IRS) on the same IBOR curve.

Hedge: Plain-vanilla IRS on the same IBOR curve, with fixed leg 30E/360; match the fixed frequency to the trades. We add the following hedging positions (per unit of the table's notional):

- Add a 10Y **payer** swap with notional 2.071474,
- Add a 5Y **receiver** swap with notional 1.000000,
- Add a 4Y **receiver** swap with notional 1.000000.

Check (under the same +1 bp parallel bump):

$$\begin{aligned}
\text{10Y payer contributes:} \quad & +\,2.071474 \times 654.853868 = +1357.000000,\\
\text{5Y receiver contributes:} \quad & -\,1.000000 \times 335.321733 = -335.321733,\\
\text{4Y receiver contributes:} \quad & -\,1.000000 \times 51.172031 = -51.172031.
\end{aligned}$$

Hedge Total:

$$+970.019227,$$

which offsets the portfolio's

$$-970.019227 \approx 0.$$

5.4 (d) Forward 4×6Y, $Z(t, T_1)$, $Z(t, T_2)$, PV01

Using the results from Question 5.a, we find:

$$Z(t, 4\text{Y}) = 0.9399942901 \quad \text{and} \quad Z(t, 6\text{Y}) = 0.9080535035.$$

The fixed-leg annuity over the period $[4\text{Y}, 6\text{Y}]$ with semi-annual coupons is:

$$\text{PV01} = 1.8397657474.$$

The floating-leg present value over $[4\text{Y}, 6\text{Y}]$ is:

$$Z(t, 4\text{Y}) - Z(t, 6\text{Y}) = 0.0319407866.$$

Dividing by the annuity gives the forward par swap rate:

$$S_{\text{fwd}}(4 \times 6\text{Y}, \text{SA}, 30\text{E}/360) = \frac{0.0319407866}{1.8397657474} = 0.01736196 \quad \text{or} \quad 1.736196\%.$$

```
Forward window: T1 = 4Y (31-OCT-2029),  T2 = 6Y (31-OCT-2031)
Z(t, T1=4Y) = 0.9400321975
Z(t, T2=6Y) = 0.9080906090
PV01 (semi-annual fixed over [T1,T2]) = 1.8398408048
Forward par swap rate S_fwd(4×6Y, SA, 30E/360) = 1.736106%
(Check) S_fwd from curve helper                = 1.736106%
```

This means that a 2-year swap starting in 4 years is fairly priced today at a fixed rate of approximately 1.736%. The discount factors decline from 0.94 at 4Y to 0.91 at 6Y, which is consistent with a modestly upward term structure over this window. The drop in Z creates a positive forward present value, and the annuity scales that into the 1.736% par rate.

Re-pricing the 4×6 swap at this fixed rate gives NPV ≈ 0, which matches the standard identity:

$$S_{\text{fwd}} = \frac{Z(4\text{Y}) - Z(6\text{Y})}{\text{PV01}} \quad \text{over } [4\text{Y}, 6\text{Y}].$$

5.5 (e) Black swaptions — Payer/Receiver, $K = 1.75\%$, vol=25%

Using Black's model on the forward 4×6 swap, the option is written on the forward swap rate under the annuity numeraire. Economically, the annuity over $[4\text{Y}, 6\text{Y}]$ scales the option values, while moneyness determines which side is more valuable. With a four-year expiry and 25% volatility, both payoffs retain meaningful time value.

On the curve, the forward rate from Question 5.d is slightly below the strike:

$$F = 1.736196\% \quad \text{vs.} \quad K = 1.75\%.$$

Therefore, the receiver swaption is slightly *in the money* and must price above the payer. That is exactly what we obtain: per unit notional,

Payer price = 0.00620885 ($\approx$ 0.620885% of notional), Receiver price = 0.00646282 ($\approx$ 0.646282% of notiona

The level of both prices is consistent with the annuity

$$A = 1.8397657474,$$

computed from the discount factors

$$Z(4\text{Y}) = 0.9399942901 \quad \text{and} \quad Z(6\text{Y}) = 0.9080535035.$$

More discounted fixed cash flows in $[4\text{Y}, 6\text{Y}]$ imply a larger scale for the option values.

```
Z(t,T1=4Y) = 0.9399942901
Z(t,T2=6Y) = 0.9080535035
PV01 (semi-annual fixed on [T1,T2]) = 1.8397657474
Forward swap rate F(4×6) = 1.736196%
Payer 4×6 swaption (K=1.75%, vol=25%)    = 0.00620885
Receiver 4×6 swaption (K=1.75%, vol=25%) = 0.00646282
```

This result is logical under Black's put–call parity for swaptions:

$$\text{Payer} - \text{Receiver} = A \times (F - K).$$

Here,

$$F - K = -0.00013804, \qquad A \times (F - K) = -0.00025399,$$

while the observed price difference is

$$0.00620885 - 0.00646282 = -0.00025397.$$

The extremely small rounding gap shows that the prices, forward rate, and annuity are mutually consistent.

5.6 (f) Swaption 1bp sensitivities and hedge

The two 4×6 swaptions are re-priced after a $+1$ bp bump to each par-swap key (2Y ... 10Y). The base prices are:

$$\text{Payer } 4 \times 6: \quad 0.00620885 \quad \text{per unit notional,}$$

$$\text{Receiver } 4 \times 6: \quad 0.00646282 \quad \text{per unit notional.}$$

```
Key (Yrs) |   Δ Payer (+1bp) |  Δ Receiver (+1bp)
-----------------------------------------------
        2 |      -0.00000005 |         0.00000007
        3 |      -0.00000008 |         0.00000011
        4 |      -0.00022780 |         0.00015917
        5 |      -0.00000123 |        -0.00000189
        6 |       0.00034115 |        -0.00023033
        7 |       0.00000000 |         0.00000000
        8 |       0.00000000 |         0.00000000
        9 |       0.00000000 |         0.00000000
       10 |       0.00000000 |         0.00000000
-----------------------------------------------
Base Payer |       0.00620885
Base Receiver |       0.00646282
```

Swaption Sensitivity and Delta Hedging:

Sensitivity sits where the forward window starts and ends. A +1 bp move at 4Y reduces the payer's value and increases the receiver's value; a +1 bp move at 6Y does the opposite. This is exactly what a 4×6 forward implies: the start-of-swap (4Y) and end-of-swap (6Y) keys drive the forward rate and the annuity, while the 5Y change is small and other keys are negligible.

Delta Hedging: Delta-hedge each option with the 4×6 forward swap on the same curve and conventions:

- Long payer swaption $\Rightarrow$ *receive-fixed* in the 4×6 forward swap. This position loses when the forward rate rises and gains when it falls, neutralizing the printed sensitivities.
- Long receiver swaption $\Rightarrow$ *pay-fixed* in the 4×6 forward swap (short receiver-forward). This offsets the receiver's opposite sensitivity pattern.

In summary, the 4Y and 6Y pillars dominate the exposure: a +1 bp at 4Y decreases the payer's value and increases the receiver's, while a +1 bp at 6Y has the reverse effect. The 5Y and other maturities contribute negligibly.

6 Question 6 — CDS: Valuation & Risk

6.1 (a) CDS curve (R=40%)

We build the *DoOrDie* CDS term structure using `FinancePy`'s `CDS` and `CDSCurve` classes, assuming a 40% recovery rate. The curve is bootstrapped with piecewise-constant hazard rates over the 1Y–5Y maturities so that each quoted spread exactly re-prices to its market input.

In each maturity interval, we solve for the hazard rate such that the present value of the premium leg equals the present value of the protection leg at the quoted spread. The survival probabilities are then obtained directly from the fitted hazard rates.

```
DoOrDie — CDS Curve from Table 3 (R = 40%)
Maturity  InputPar_bp  RepricedPar_bp  SurvivalProb  Hazard_1perYear
      1Y           70            70.0      0.988225         0.011845
      2Y           74            74.0      0.975340         0.013125
      3Y           80            80.0      0.960406         0.015430
      4Y           85            85.0      0.944238         0.016978
      5Y           88            88.0      0.928202         0.017129
```

The re-pricing column matches the inputs at each maturity, confirming that the curve is fitted exactly to the quoted CDS spreads. The implied hazard rate is gently upward-sloping, producing a declining survival curve from 98.82% at 1Y to 92.82% at 5Y.

The magnitudes are consistent with the course's back-of-the-envelope relation:

$$\text{spread} \approx \text{hazard} \times (1 - \text{recovery}),$$

so that, for 1Y,

$$\frac{70 \text{ bp}}{1 - 0.40} \approx 1.17\%.$$

6.2 (b) Market spread for 3.5Y (maturity 20/03/2029)

Using the CDS curve built in Section 6.a for *DoOrDie* (recovery rate 40%, piecewise-constant hazard rates fitted to the 1Y–5Y quotes), I price the market par spread at an explicit maturity of 20-Mar-2029 with an effective date of 20-Sep-2025.

```
CDS — Market Par Spread at Explicit Maturity
----------------------------------------
Effective date : 20-SEP-2025
Maturity date  : 20-MAR-2029
Recovery (R)   : 40%
Par spread     : 81.50 bp
```

The target date lies between the 3Y and 4Y maturities, so it naturally falls between 80 and 85 bps. The fitted hazard rates from Section 6.a imply a smooth decline in survival probabilities, making a par spread of approximately 81.5 bps fully consistent with the curve.

6.3 (c) Value of a long protection (spread 120bp, mat. 21/03/2029, notional $20m)

We mark to market the existing long-protection CDS on the issuer curve built in Section 6.a. The contract runs at a spread of 120 bps, while the current market spread for the same dates is 82.82 bps.

The value is computed as:

$$\text{Value} = \text{PV(protection leg at today's curve)} - \text{PV(premium leg at the contract spread)}.$$

Equivalently, the value per unit notional can be approximated as:

$$\text{Value} \approx (\text{par spread} - \text{contract spread}) \times \text{risky annuity},$$

where the *risky annuity* is the present value of one basis point of running premium.

```
Q6(c) — Long protection CDS valuation
-------------------------------------
Start (settlement) : 20-SEP-2025
Maturity           : 21-MAR-2029  (~42 months)
Recovery (R)       : 40%
Contract spread    : 120.00 bp
Par spread (market): 81.32 bp

Clean PV per unit  : -0.013148
Total PV on $20m   : -262,966.05
```

The position is *out of the money*: you are paying 120 bps while the market would pay 82.8 bps for identical protection, so the mark is negative

$$-\$263{,}000 \text{ on a } \$20{,}000{,}000 \text{ notional.}$$

6.4 (d) Sensitivity to $R \in \{0, 10, 20, 30\}\%$

We re-priced the same long-protection CDS while varying the recovery rate R and re-bootstrapping the issuer curve each time to the same market quotes (Table 3).

```
Valuation date (curve): 20-SEP-2025  |  Settlement: 20-SEP-2025
Q6(d) — Long protection CDS @ 120 bp (PV vs Recovery)
Recovery R  Par spread (bp)   PV total ($)
        0%            81.32 -265276.975120
       10%            81.32 -264889.969966
       20%            81.32 -264407.257788
       30%            81.32 -263788.327983
```

From $R = 0\%$ to $R = 30\%$, the present value (PV) improves by about \$1,488.65, roughly 0.56% of the $\sim$ \$265,000 mark-to-market. This change is not economically significant.

Why the PV barely moves.

- Re-bootstrap keeps market par fixed. For each recovery rate R, the issuer curve is re-fitted so that the quoted CDS still price exactly at par. The implied hazard rate λ adjusts according to the approximate relation

$$s \approx (1 - R)\lambda.$$

When R increases, λ increases as well, keeping the par spreads essentially unchanged.

- Off-market mechanics dominate. The trade pays 120 bps while the market (par) level is about 82.83 bps, so the mark-to-market value is driven by

$$\text{PV} \approx (\text{par} - \text{contract}) \times \text{risky annuity} \times \text{notional}.$$

After each re-bootstrap, the risky annuity changes only slightly with R, hence the PV shifts only marginally.

- Residual effects only. The small remaining variation arises from implementation details: accrual-on-default, exact coupon timing, and the re-implied shape of the hazard curve between maturities.

6.5 (e) DV01 by bucket (1–5Y) and dealer hedge

We compute the trade's credit-spread key sensitivities by bumping the CDS par spread at each quoted node (+1 bp at 1Y…5Y), re-bootstrapping the issuer curve each time, and re-pricing the 3.5-year CDS.

MaturityY	ΔPV(+1bp) [currency]	ΔPV(+1bp) [bps of N]
1	11.983980	0.005992
2	24.101127	0.012051
3	2958.785561	1.479393
4	3917.904903	1.958952
5	0.000000	0.000000

Base 3.5Y par spread (bp): 82.858 | Risky annuity A: 3.416948

Hedging Strategy: Sell protection at 3Y and sell protection at 4Y, as this is where the risk lies (the largest sensitivities). We choose the hedge notionals such that, at each node,

$$\text{hedge value change} \approx -\,\text{trade value change}.$$

In other words:

- Sell protection on the 3Y CDS to hedge the +1 bp value change at 3Y.
- Sell protection on the 4Y CDS to hedge the +1 bp value change at 4Y.

References

- Course notes, `FinancePy` docs, and cited materials.

Appendices

- `ADVANCED_DERIVATIVES_COURSEWORK_85769_85842_85944.ipynb`